Week 8 Solana Vault Card

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The official Week 8 vault-card assignment is complete. All three extensions were implemented and verified: blockchain creation time, owner-only closure with rent refund, and administrative pause of new cards. The program is deployed on Solana Devnet.

Implementation

The instructor's initialize_vault and create_user_card structure is retained. VaultState is a PDA derived from the token mint; it stores admin, mint/account links, bump and paused. It is the authority for the share mint and token vault. SPL account data is owned by the Token Program. UserCard is derived from vault and user and stores vault, owner, bump and created_at.

Requirement	Implementation and verification
Creation time	Clock::get()?.unix_timestamp; card space is 81 bytes. Timestamp and owner/vault links are checked.
Close and refund	Canonical PDA and owner signature required. close=user returns rent. Closure is permitted while paused.
Administrative pause	VaultState.paused starts false. Only admin can call set_paused. A new user receives VaultPaused; failed creation leaves no card.

Verification results

Anchor 0.31.1 build, IDL generation and TypeScript checking passed. All 16 validator integration tests passed. Negative cases cover duplicate creation, forged PDA, missing signature, non-admin pause, unauthorized closure, wrong vault and direct share mint by admin. State preservation and account removal are checked.

Devnet confirmed initialization, card creation, pause, closure, resume and recreation. Paused creation by a new user was rejected. Closure returned 1,061,720 lamports; after the 5,000-lamport fee, the owner's balance increased by 1,056,720. The card was absent after closure and recreated after resume. Final paused=false, share supply=0 and token-vault balance=0.

The deployed executable matches the local binary, with zero padding in unused ProgramData allocation. Evidence includes transaction slots, raw RPC account data, binary hashes and actual test logs. No deposits, withdrawals or Raydium operations are included in Week 8.

Repository and Explorer evidence

GitHub project | Instructor starter

Deployed program

initializeVault | createUserCard | pause

closeUserCard | resume | recreateUserCard